

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

INTERNATIONAL GCSE

BIOLOGY

Paper 1

Wednesday 9 May 2018 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



Answer **all** questions in the spaces provided.

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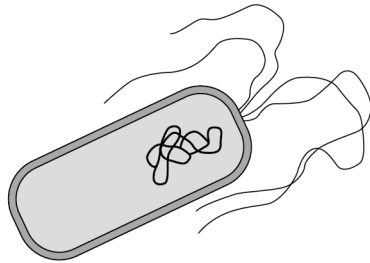
0 1

Organisms are made from cells.

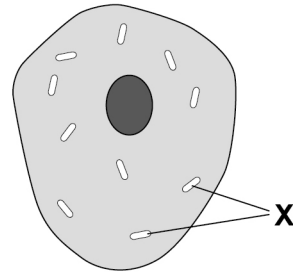
Figure 1 shows two types of cell.

Figure 1

Bacterial cell



Animal cell



Not to scale

0 1 . 1

Complete **Table 1**.

Use information from **Figure 1**.

Tick each box if the cell has that part.

[4 marks]

Table 1

Cell part	Bacterial cell	Animal cell
Cell wall		
Nucleus		
Cell membrane		
Cytoplasm		



0 1 . 2

The structures labelled **X** in **Figure 1** are where most of the energy is released in respiration.

What is the name of **X**?

[1 mark]

Tick **one** box.

Chloroplasts

☐

Mitochondria

☐

Plasmids

☐

Ribosomes

☐**0 1 . 3**

Name **one** structure that is in a plant cell but **not** in an animal cell.

[1 mark]

0 1 . 4

Give the function of the structure you named in **01.3**.

[1 mark]

Question 1 continues on the next page

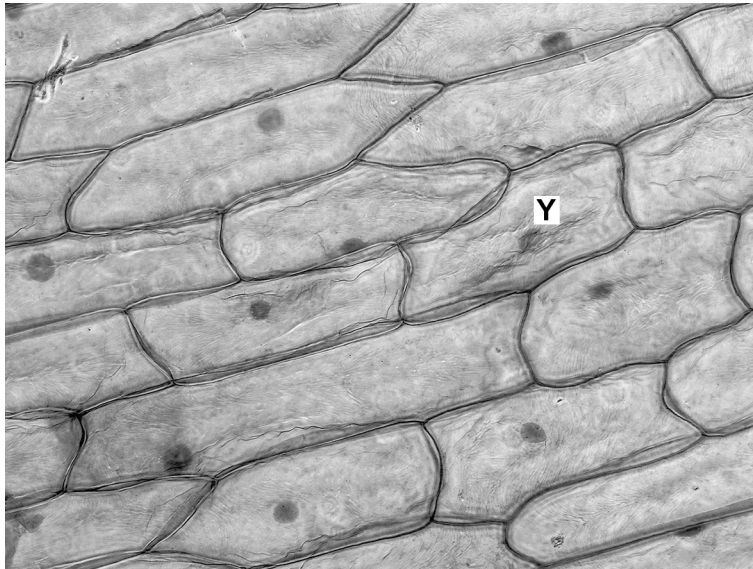
Turn over ►



0 1 . 5

Figure 2 shows plant cells seen through a microscope.

Figure 2



In **Figure 2** the image width of the plant cell labelled **Y** is 20 mm

The plant cell has been magnified 1000 times.

Calculate the actual width of the plant cell.

Use the equation:

$$\text{actual width} = \frac{\text{image width}}{\text{magnification}}$$

[2 marks]

actual width = _____ mm

9



0 2

Animals have different adaptations to help them survive in their environments.

0 2 . 1

Figure 3 shows a lizard called a thorny devil.

Figure 3



Look at **Figure 3**. Give **one** adaptation of the thorny devil.

Suggest how this adaptation is an advantage to help the thorny devil survive.

[2 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 2

Figure 4 shows a fennec fox. The fennec fox lives in the desert and hunts at night.

Figure 4



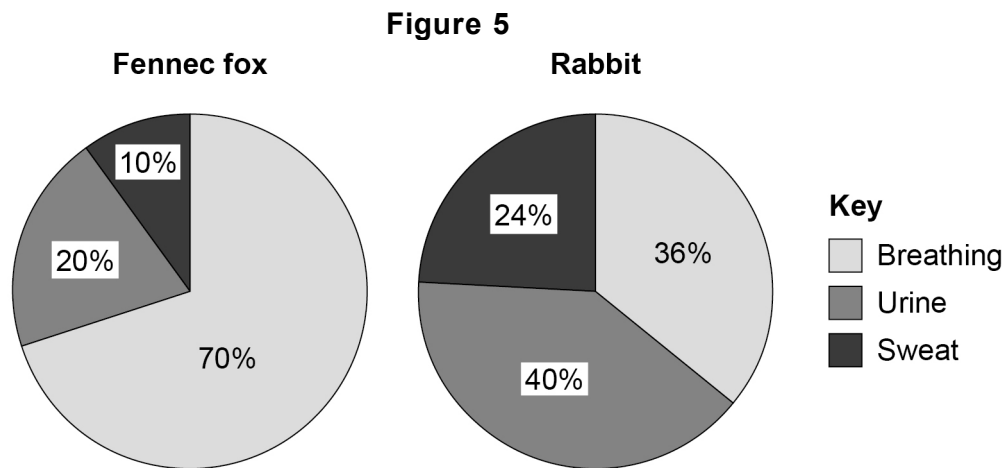
Look at **Figure 4**. Give **one** adaptation of the fennec fox.

Suggest how this adaptation is an advantage to help the fennec fox survive.

[2 marks]

0 2 . 3

Figure 5 shows the different ways that water is lost by a fennec fox and by a rabbit.



What proportion of water is lost in urine by the fennec fox compared to the rabbit?

[1 mark]

Tick **one** box.

half

☐

2 times

☐

quarter

☐

4 times

☐

0 2 . 4

Which organ detects changes in the water content of the blood?

[1 mark]

Tick **one** box.

Brain

☐

Kidney

☐

Lung

☐

Skin

☐

Turn over ►



0 2 . 5

Name the hormone that controls the water content of the blood.

[1 mark]

0 2 . 6

Most water in cells comes from food and drink.

Some chemical reactions in the body also produce water.

Name **one** chemical reaction in the body that produces water.

[1 mark]

0 2 . 7

Urine contains the products from the breakdown of excess amino acids.

Describe how the liver breaks down amino acids for removal from the body.

[3 marks]



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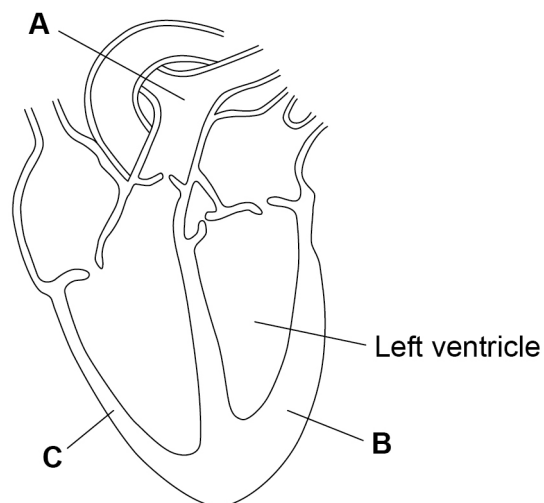


0 3 . 1

Blood contains several components.

Draw **one** line from each component to its function.**[3 marks]**

Component	Function
Plasma	carries oxygen
Platelet	helps to destroy pathogens
White blood cell	contains water and proteins
	helps the blood to clot

Figure 6 shows a section through a human heart.**Figure 6**

0 3 . 2

Name the blood vessel **A**.**[1 mark]**



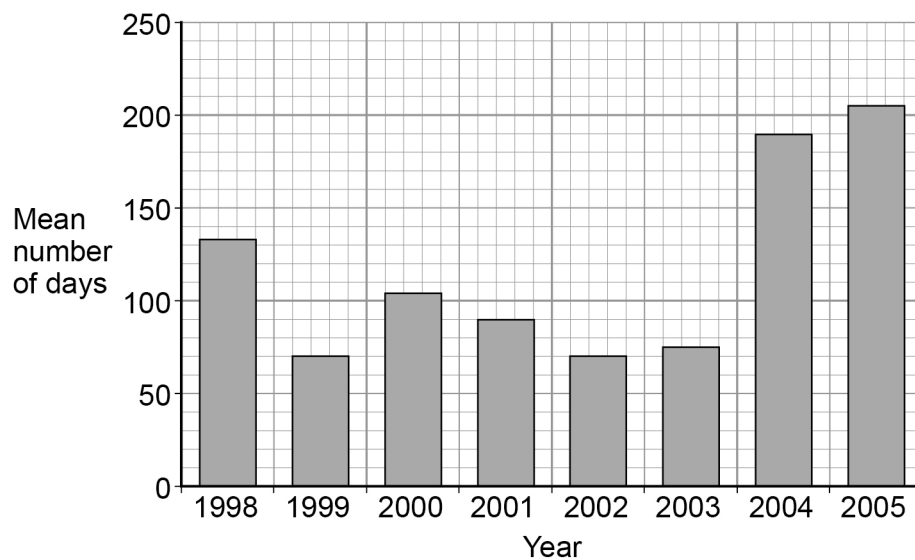
0 3 . 3

Explain why the wall of the heart is thicker at **B** than at **C**.**[3 marks]**

0 3 . 4

People with heart disease may receive a heart transplant.

Figure 7 shows the mean number of days people with heart disease waited for a heart transplant in one country between 1998 and 2005.

Figure 7

Calculate the difference between the mean number of days people waited for a heart transplant in 2005 compared with 2002.

[1 mark]

Difference = _____ days

Turn over ►

14



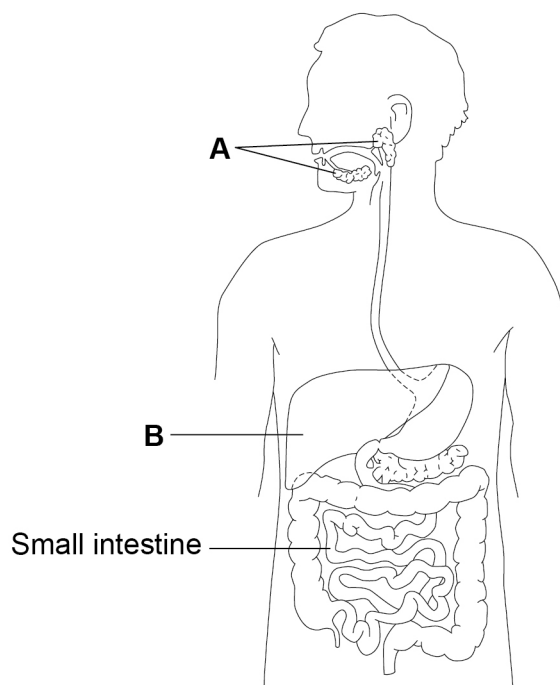
0 4

Enzymes are biological catalysts.

Enzymes in the digestive system break down food molecules.

Figure 8 shows the digestive system.

Figure 8

**0 4 . 1**

Name organs **A** and **B**.

[2 marks]

A _____

B _____

Question 4 continues on the next page

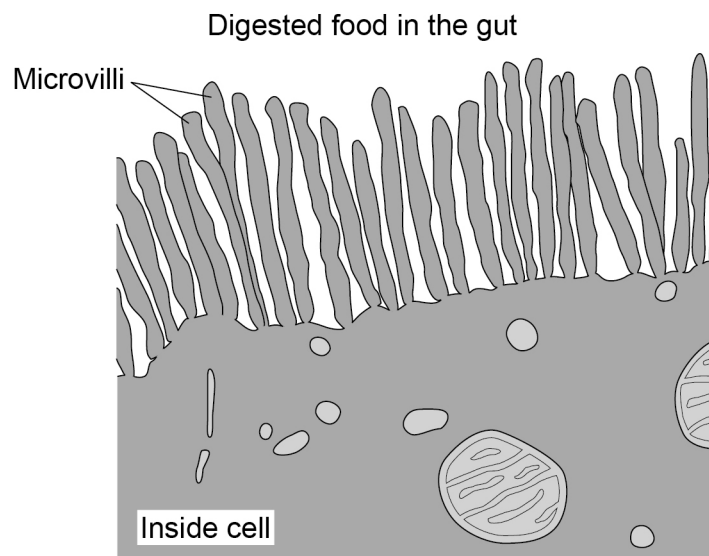
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0 4 . 2

Figure 9 shows part of a cell lining the small intestine.

Figure 9



Explain how microvilli help digested food to be absorbed into the cell.

Use information from **Figure 9**.

[2 marks]

0 4 . 3

Lipase is an enzyme that breaks down lipid.

Complete the equation.

[1 mark]



0 4 . 4

Describe **one** role of bile in digestion.**[2 marks]**

0 4 . 5

A student investigated the effect of temperature on the time taken for lipase to digest a lipid.

This is the method used.

1. Mix some lipid and lipase in a test tube.
2. Place the test tube in a water bath at 25 °C.
3. Record the time taken for the solution to reach pH5, indicating that all the lipid has been digested.
4. Repeat steps 1 to 3 at different temperatures.

The teacher said the investigation was **not** valid.Suggest **two** improvements the student could make to their investigation to make it more valid.**[2 marks]**

1

2

Question 4 continues on the next page**Turn over ►**

The student then carried out a second investigation using a valid method.

Table 2 shows the results.

Table 2

Temperature in °C	Time taken to digest lipid in seconds			
	Trial 1	Trial 2	Trial 3	Mean
20	491	545	554	530
30	468	241	482	X
40	367	380	363	370
50	389	407	359	385
60	no result	no result	no result	no result

0 4 . 6

Calculate value **X** in **Table 2**.

[2 marks]

X = _____ s



0 4 . 7

The student made this conclusion.

'The best temperature for lipase is 40 °C.'

Suggest why this conclusion may not be correct.

[2 marks]

0 4 . 8

The solution at 60 °C did not reach pH5 after 900 seconds.

Give the reason why.

[1 mark]

14

Turn over for the next question**Turn over ►**

0 5

Retinoblastoma is one of the most common cancers in children.

It causes malignant tumours to develop in the retina of the eye.

0 5 . 1

What is a malignant tumour?

[3 marks]



Light receptors are found in the retina of the eye.

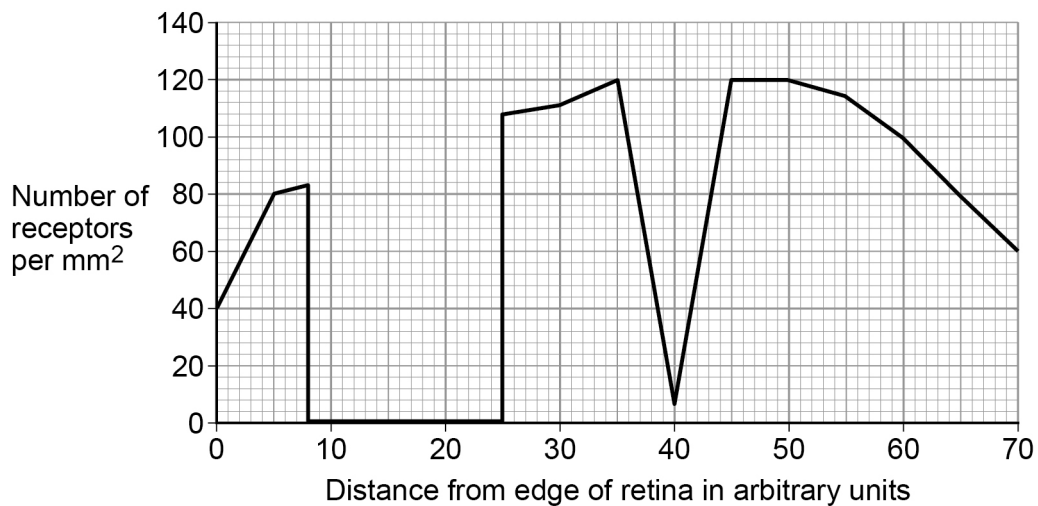
Figure 10 shows the number of receptors across a section of a retina in a child without a tumour.

Figure 10
Retina without tumour



Figure 11 shows the number of receptors across a section of a retina in a child with a tumour.

Figure 11
Retina with tumour



0 5 . 2

Describe the effect the tumour has on the receptors in the eye.

Use data from **Figure 10** and **Figure 11**.

[1 mark]

Turn over ►



0 5 . 3

Explain why the changes to the receptors seen in **Figure 10** and **Figure 11** might affect the child's vision.

[2 marks]



Some forms of retinoblastoma are genetically inherited.

The allele for retinoblastoma is recessive.

0 5 . 4

What does recessive mean?

[1 mark]

0 5 . 5

A man and a woman have a child who developed retinoblastoma.

Neither parent has retinoblastoma.

Draw a genetic diagram to show the probability that the parents' next child will develop retinoblastoma.

You should identify any of the children with retinoblastoma.

Use the following symbols:

R = allele for not having retinoblastoma

r = allele for retinoblastoma

[4 marks]

Probability = _____

11

Turn over ►

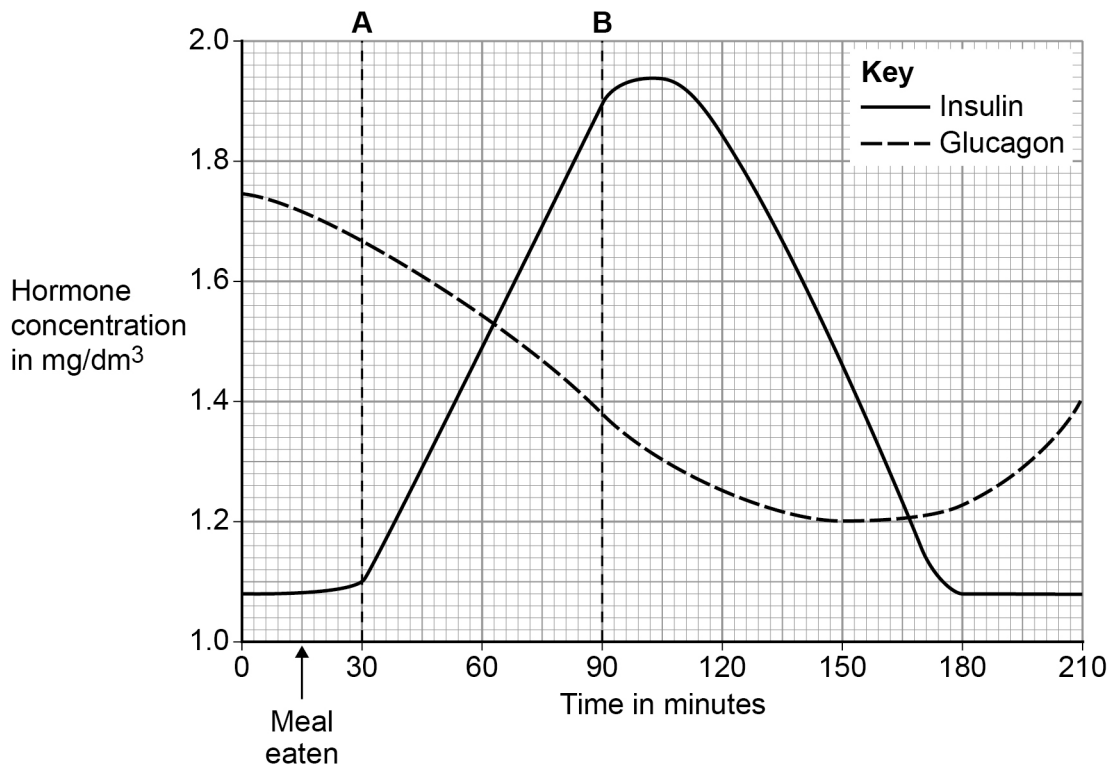


0 6

The human body needs to control blood glucose concentration within narrow limits.

Figure 12 shows how the concentrations of insulin and glucagon change after eating a meal.

Figure 12

**0 6 . 1**

Name the organ that produces glucagon.

[1 mark]

0 6 . 2

Describe how the release of insulin changes blood glucose levels.

[2 marks]



0 6 . 3

Calculate the rate of insulin release into the blood between points **A** and **B** on **Figure 12**.

You should include the units in your answer.

[3 marks]

Rate of insulin release = _____ Unit _____

Question 6 continues on the next page

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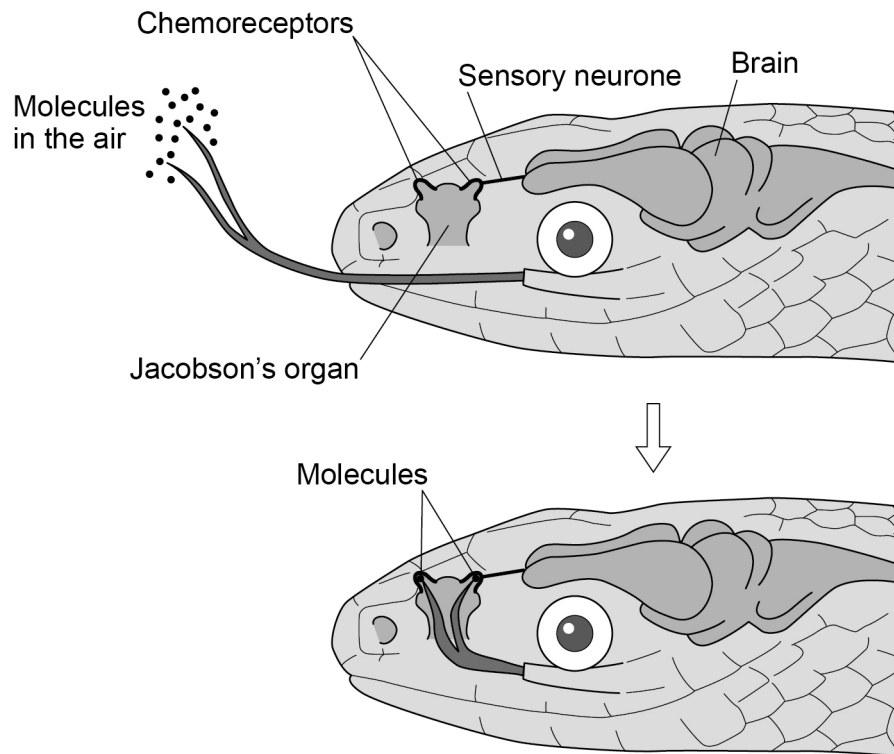
07

Snakes are almost blind. They have a specialised structure called the Jacobson's organ to help them detect molecules in the air. This helps snakes to detect prey.

The Jacobson's organ is located in the roof of the snake's mouth.

Figure 14 shows how the tongue brings the molecules to the Jacobson's organ.

Figure 14



07.1

Suggest the role of chemoreceptors.

[1 mark]

07.2

Suggest **two** benefits for the snake of having a forked tongue.

[2 marks]

1

2



0 7 . 3

The pain withdrawal reflex action is important in humans. A finger will be pulled away rapidly after touching a hot surface.

Describe how the pathway from the Jacobson's organ to the brain is different from the pathway taken during the pain withdrawal reflex.

Use information from **Figure 14** and your own knowledge.

[2 marks]

0 7 . 4

Suggest **one** reason why it is important that a reflex action is rapid.

[1 mark]

Question 7 continues on the next page

Turn over ►

Some snakes produce poisons which affect synapses.

0 7 . 5

Describe what happens when an impulse reaches a synapse.

[2 marks]

0 7 . 6

The poison prevents an impulse being passed across the synapse.

Explain why people bitten by these snakes find it hard to inhale.

[4 marks]

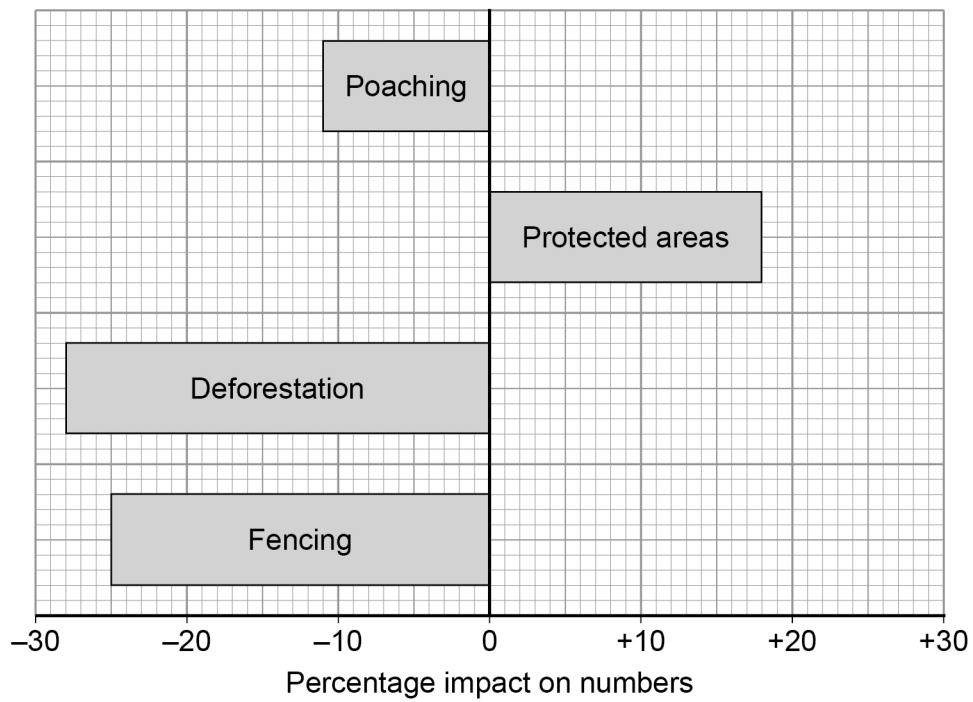
12



08

Figure 15 shows how different human activities affect the number of blue wildebeest in the wild.

Figure 15



08.1

Describe the relative effects of human activities on blue wildebeest numbers.

[3 marks]

Question 8 continues on the next page

Turn over ►



0 8 . 2

Over 1 million years ago the only wildebeest was the blue wildebeest.

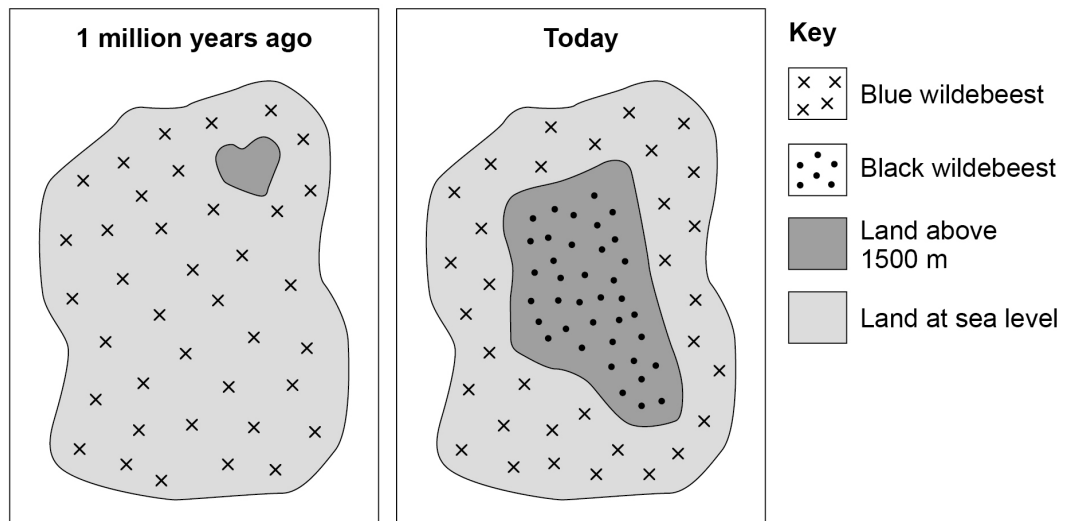
Over time the land where wildebeest lived changed. Some land rose high above sea level and is known as the Highveld.

The wildebeest must be on the same level as their partner to mate.

There are now two species: the original blue wildebeest and a newer species known as black wildebeest.

Figure 16 shows the distribution of wildebeest species 1 million years ago and today.

Figure 16



Explain how the development of the Highveld led to the formation of two separate species of wildebeest.

[5 marks]

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END OF QUESTIONS



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